

RESEARCH PROFILE

PhD researcher developing and applying statistical methods for incomplete, functional, longitudinal, high-dimensional, and time-to-event biomedical data. Experience spans method development, simulation and high-performance computing, survival analysis, predictive modeling, biomarker analytics, and collaborative study design.

EDUCATION

George Washington University, Milken Institute School of Public Health

PhD in Biomedical and Health Data Science, Biostatistics track

Aug 2024-present

New York University, School of Global Public Health

MS in Biostatistics

Sep 2022-Jul 2024

SELECTED EXPERIENCE

Part-time Biostatistician, DPPOS Study | GWU Biostatistics Center

- Develop functional-principal-component-based multiple-imputation methods for incomplete functional and scalar data; evaluate performance through large-scale simulation and GW Pegasus high-performance computing.
- Lead statistical analysis plan development for longitudinal proteomic analyses of MCI and AD/ADRD biomarkers in type 2 diabetes.
- Evaluate extracellular-vesicle biomarker quality control and its impact on downstream analyses of neuronal and systemic insulin-signaling measures.

Oct 2024-present

Part-time Biostatistical Consultant | BC3, New York University

- Collaborated with the Barth Syndrome Foundation and NYU Langone to co-lead statistical design and analysis of survival and clinical progression in Barth syndrome.
- Served as primary biostatistician for evaluation of the NICHE aging-care implementation model.

Jan 2024-Nov 2025

Course Assistant | New York University

- Supported Applied Survival Analysis, Linear Regression and Modeling, Data Management and Statistical Computing, and Biostatistics for Public Health.

Sep 2023-May 2024

Graduate Student Mentor | NYU PQAR Summer Program

- Guided undergraduate researchers through aging-related statistical design, analysis, and presentation.

Jun-Jul 2023

Business Analyst Intern | Baidu AI Cloud, Intelligent Healthcare Unit

- Supported solution design for electronic medical-record quality control, data management, clinical decision support, patient services, and hospital operations.

Jul-Nov 2022

TECHNICAL SKILLS

Programming: R, Python, SQL, SAS | **Computing and research tools:** HPC Pegasus, AWS, Google Colab, EAST, ArcGIS Pro

PEER-REVIEWED PUBLICATIONS

1. **Fu K**, Huang Y, Bowen V, et al. Survival and Clinical Progression in Barth Syndrome: Insights From the Barth Syndrome Foundation's Database of 502 Affected Individuals. *Journal of Inherited Metabolic Disease*. 2026. doi:10.1002/jimd.70189
2. Zhao H, Zhou X, **Fu K**, et al. Prospective cohort study evaluating the association between influenza vaccination and neurodegenerative diseases. *npj Vaccines*. 2024;9:51. doi:10.1038/s41541-024-00841-z
3. Wan L, Geng F, **Fu K**, Chen W. Low-intensity pulsed ultrasound in postoperative fracture healing: Current research landscape and hotspot analysis. *Asian Journal of Surgery*. 2025;48(10):6114-6116. doi:10.1016/j.asjsur.2025.06.198

SELECTED PRESENTATIONS

- Functional Principal Components-Based Multiple Imputation for Functional Data - JSM contributed paper, 2026.
- Functional Regression-Based Multiple Imputation for Mixed Missing Data - GWSPH Research Day poster, 2026.
- Comparison of Surgical and Obstetric Outcomes Between Two Types of Cervical Polypectomy During Pregnancy - ENAR poster, 2025.
- Evaluating Feasibility and Adaptability of Stronger Together Peer Mentoring Program for Cancer Patients in Viet Nam Using CFIR - IPOS poster, 2024.
- Evaluating the NICHE Program: Disparities in Aging Care Implementation Across U.S. Hospitals - Howard-Meyers Bernadine Lacey Health Equity conference poster, 2024.

SELECTED HONORS

- 2026** GWSPH Research Day - Biostatistics & Bioinformatics Prize, Doctoral Research
- 2026** ENAR DataFest Winner - GWU team leader
- 2024** NYU Thesis Recognition Award - Innovative Topic or Approach
- 2024** NYU GPH Biostatistics Community Recognition Award
- 2019** National Scholarship, China

LEADERSHIP AND SERVICE

HDS Program Student Representative, GWU (2026-present) | ASA Biopharmaceutical Section RISW Volunteer (2025) | ICSA Volunteer (2026)